

MSc. in Computing Practicum Approval Form

Section 1: Student Details

Project Title:	Dual-Stream Vision Transformer for Multi-Domain Skin Disease Classification Using Frozen DINOv2
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Chosen major:	Data Analytics
Supervisor	Luca Rossetto
Date of Submission	

Section 2: About your Practicum

What is the topic of your proposed practicum? (100 words)

This practicum develops a dual stream deep learning system for unified skin disease classification across three distinct medical domains: oncological lesions (melanoma, basal cell carcinoma, squamous cell carcinoma), benign conditions (nevi, keratosis, dermatofibroma, vascular lesions), and infectious diseases (chickenpox, monkeypox, measles, HFMD). Using a frozen DINOv2 Vision Transformer backbone with domain-aware architecture, the research investigates whether explicit separation of oncological and infectious disease processing improves classification accuracy across 14 fine-grained disease types. The approach addresses severe class imbalance (54:1 ratio) through focal loss and contrastive learning. With 99% parameter efficiency via frozen backbone, the system enables resource-constrained deployment while maintaining competitive diagnostic performance suitable for automated screening applications.

Please provide details of the papers you have read on this topic (details of 5 papers expected).

[1] DINOv2: Learning Robust Visual Features without Supervision - Oquab et al. [1] introduce DINOv2, a self-supervised Vision Transformer achieving state-of-the-art performance across image classification, segmentation, and retrieval without fine-tuning. Trained on 142 million curated images, DINOv2 produces frozen features matching weakly-supervised models while eliminating text dependency, demonstrating strong transfer learning for medical imaging with limited labels.

[2] A Deep CNN Transformer Hybrid Model for Skin Lesion Classification - Nie et al. [2] combine ResNet-50 with Vision Transformer for ISIC 2018 classification. Their CNN-ViT hybrid with focal loss addresses 58:1 class imbalance, improving accuracy from 74% to 89% with 0.90 macro-AUC, demonstrating transformer-based attention enhances dermoscopic classification.

[3] Focal Loss for Dense Object Detection - Lin et al. [3] propose focal loss to address extreme class imbalance by down-weighting easy examples and focusing on hard negatives. Applied to RetinaNet, focal loss achieves 39.1 AP on COCO, demonstrating effectiveness for imbalanced datasets applicable to medical imaging.

[4] Unified Multi-Task Learning Model - Al-masni et al. [4] present multi-task framework for simultaneous segmentation and classification, achieving 82.07% F1-score versus 78.28% baseline on ISIC 2016. Joint reverse optimization prevents task dominance, demonstrating correlated tasks mutually enhance performance through shared representations.

[5] Monkeypox Skin Lesion Detection - Ali et al. [5] create first monkeypox dataset (228 images), achieving 82.96% accuracy using ResNet50 for monkeypox versus chickenpox/measles discrimination. This study highlights challenges distinguishing visually similar infectious diseases and establishes baseline performance for pox virus classification.

How does your proposal relate to existing work on this topic described in these papers? (200 words)

This proposal extends existing research by addressing a gap in unified multi-domain classification. While [1] establishes DINOv2's transfer learning capabilities and [2] demonstrates Vision Transformers for dermatoscopic lesions, no prior work unifies oncological and infectious disease classification. Existing studies treat these as separate problems [2] focuses exclusively on melanoma/nevus distinction, while [5] addresses infectious disease independently.

The proposed dual-stream architecture extends multi-task learning from [4] beyond segmentation-classification to domain-specific processing. Unlike [2]'s single-stream on homogeneous ISIC 2018 data, this work tackles 14 heterogeneous classes spanning three medical domains with 54:1 imbalance. Focal loss from [3] provides the imbalance-handling foundation, augmented with contrastive learning for rare classes. Work from [5] informs infectious disease stream design, extending beyond binary monkeypox detection to multi-disease classification.

The three-tier categorization (oncological/benign/infectious) with 14 fine-grained subclasses represents a novel taxonomy absent in [1-5]. The frozen backbone approach (99% parameter efficiency) builds on [1]'s transfer learning while [4]'s joint optimization inspires the dual-domain architecture. This unified framework enables single-model deployment for diverse screening scenarios, from melanoma detection to outbreak surveillance.

What are the research questions that you will attempt to answer? (200 words)

Main Research Question: Can a frozen DINOv2 Vision Transformer with dual-stream architecture accurately classify skin diseases across three distinct medical domains (oncological, benign, infectious) while handling severe class imbalance?

Sub-Research Question: Does domain-aware dual-stream processing improve fine-grained classification accuracy for the 14 disease subclasses compared to single-stream baseline, particularly for minority classes with fewer than 1000 training samples?

How will you explore these questions? (Please address the following points. Note that three or four sentences on each will suffice.)

1. Software Environment:

Development uses Python 3.10 with PyTorch 2.0+ in Google Colab notebooks utilizing T4 or L4 GPUs (16GB VRAM). Key libraries include Transformers (Hugging Face) for DINOv2, Timm for vision utilities, Albumentations for augmentation, Weights & Biases for tracking, Scikit-learn for metrics, and Matplotlib/Seaborn for visualization.

2. Development Work:

Phase 1 - Data Pipeline: Implementation of custom PyTorch Dataset handling 14-class labels with three-domain grouping (oncological/benign/infectious). Class-specific augmentation applies heavy transformations for classes with fewer than 1000 samples and moderate augmentation for majority classes. Balanced sampling ensures each domain receives proportional representation during training.

Phase 2 - Model Architecture: Construction of frozen DINOv2-ViT-B/16 backbone (86M parameters) with trainable classification heads. Dual-stream design includes oncological stream (processing melanoma, BCC, SCC, actinic keratosis, nevi, keratosis, dermatofibroma, vascular) and infectious stream (processing monkeypox, chickenpox, cowpox, HFMD, measles, healthy). Domain routing module learns soft weights determining stream activation. Final classification combines both streams through weighted fusion producing 14-class predictions.

Phase 3 - Training Framework: Multi-task loss combines focal loss for class imbalance, domain routing loss for stream specialization, and supervised contrastive loss for rare class embeddings. Training uses AdamW optimizer with learning rate 1e-4, batch size 32, and early stopping on validation F1-score. Total trainable parameters: approximately 500K (0.6% of total), with DINOv2 backbone completely frozen.

Phase 4 - Evaluation Suite: Development of comprehensive metrics including per-class precision/recall/F1, three-domain accuracy, 14-class confusion matrices, and GradCAM attention visualization showing model focus regions for different disease types.

3. Data Availability:

Data immediately available via Hugging Face: `datasets.load_dataset("ahmed-ai/skin-lesions-classification-dataset")`. Pre-split into train (29,322), validation (3,660), test (3,674) with stratified

distributions. Preprocessing includes quality filtering, three-domain label generation, class weight computation for focal loss, and augmentation strategy assignment based on class size.

4.Experiments:

Experiment 1 - Baseline: Train frozen DINOv2 with single linear classifier (768→14) using standard cross-entropy to establish baseline performance across three domains and 14 subclasses.

Experiment 2 - Dual-Stream Architecture: Implement full dual-stream with domain routing, comparing against single-stream baseline on three-domain accuracy, 14-class F1-scores, and domain-specific performance metrics.

5.Expected Output:

Quantitative Metrics: Overall test accuracy and macro/weighted F1-scores across 14 classes. Per-class precision, recall, F1-score tables. Three-domain classification accuracy (oncological/benign/infectious). 14-class confusion matrices. Minority class recall measurements. Inference latency (target: under 100ms per image on T4 GPU). Memory footprint analysis.